THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM ONE
033 BIOLOGY
MARKING SCHEME AUGUST 2026

SECTION A: 15 MARKS

1. **10 marks**

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	B	B	C	C	C	B	C	C	B

2. **5 mark**

(i)	(ii)	(iii)	(iv)	(v)
C	A	D	B	E

SECTION B: 70 MARKS

3. (a) Two apparatus/resources used to obtain measurements — **2 marks**

Any two:

(i) Ruler

(ii) Balance

(iii) Measuring cylinder, if appropriately used

(iv) Any other suitable measuring apparatus from the scenario

(b) Laboratory safety precautions — **2 marks**

Any two:

(i) Handle apparatus carefully.

(ii) Keep the working area clean.

(iii) Do not eat or drink in the laboratory.

(iv) Follow the teacher's instructions.

(v) Report accidents immediately.

(vi) Wash hands after the experiment.

(c) Suitable hypothesis — **2 marks**

Example:

Increasing the amount of water supplied to bean seedlings increases their growth up to an appropriate level.

(d) Variables — **4 marks**

Variable	Expected answer
(i) Independent variable	Amount/quantity of water supplied
(ii) Dependent variable	Growth of bean seedlings, e.g. height

(iii) Controlled variable	Any one: type of seed, soil, container, amount of sunlight, temperature, etc.
(iv) Control group condition	Seedlings receiving the normal/standard amount of water for comparison

Total = 10 marks

4. (a) Meaning of Biology — **2 marks**

Biology is the branch of science that deals with the study of living organisms.

(b) Importance of studying Biology **— 4 marks**

Any four:

(i) Helps in understanding living organisms.

(ii) Helps in disease prevention and control.

(iii) Improves agricultural practices.

(iv) Helps in environmental conservation.

(v) Provides knowledge about nutrition and health.

(vi) Provides knowledge useful in medicine.

(vii) Creates employment/career opportunities.

(viii) Helps solve biological problems in society.

(c) Related scientific fields and contributions **— 4 marks**

Any two.

(i) **Chemistry** — helps explain chemical substances and reactions occurring in living organisms.

(ii) **Physics** — helps explain physical processes such as movement, energy and forces in organisms.

(iii) **Geography** — helps study the distribution of organisms and their relationship with the environment.

(iv) **Mathematics/Statistics** — helps in collecting, analysing and interpreting biological data.

Total = 10 marks

5. (a) Identification and reason — **2 marks**

Specimen **X: Plant cell**

Reason: It contains structures such as:

- Cell wall
- Chloroplasts
- Large permanent vacuole

(b) Four differences between plant and animal cells **— 4 marks**

Plant cell	Animal cell
(i) Has a cell wall	(i) Lacks a cell wall
(ii) Usually has chloroplasts	(iii) Lacks chloroplasts

(iv) Has a large permanent vacuole	(v) Vacuoles are absent or small/temporary
(vi) Usually has a more regular/fixed shape	(vii) Usually has an irregular/flexible shape

(c) Importance of cell differentiation **— 4 marks**

Any two well-explained points:

(i) Allows cells to become specialized for particular functions.

(ii) Enables formation of tissues and organs.

(iii) Improves efficiency of different body functions.

(iv) Makes division of labour possible in multicellular organisms.

Total = 10 marks

6. (a) Kingdoms — **4 marks**

Organism	**Kingdom**
P	**Monera**
Q	**Plantae**
R	**Fungi**
S	**Animalia**

(b) Reasons for classification **— 3 marks**

Any three:

(i) Makes identification of organisms easier.

(ii) Makes study of organisms easier.

(iii) Shows similarities and differences among organisms.

(iv) Shows evolutionary/relationship patterns.

(v) Makes communication about organisms easier.

(vi) Organizes the large number of living organisms into manageable groups.

(c) Binomial nomenclature **— 1 mark**

A system of giving each organism a scientific name consisting of **two names**, the genus and species.

(d) Advantages of scientific names **— 2 marks**

Award **1 mark each** for any two:

(i) They are universally accepted.

(ii) Avoid confusion caused by different common names.

(iii) Each organism has a unique scientific name.

(iv) Facilitate communication among scientists worldwide.

Total = 10 marks

7. (a) Definition of a virus — **2 marks**

A virus is a microscopic infectious agent consisting mainly of genetic material surrounded by a protein coat and which can reproduce only inside a living host cell.

(b) Three characteristics of viruses **— 3 marks**

Any three:

(i) Very small/microscopic.

(ii) Have genetic material.

(iii) Have a protein coat/capsid.

(iv) Cannot reproduce independently.

(v) Reproduce inside living host cells.

(vi) Can cause diseases.

(vii) Show both living and non-living characteristics.

(c) Two differences between viruses and bacteria — **2 marks**

Any two valid differences.

Viruses	Bacteria
Acellular	Cellular
Cannot reproduce independently	Can reproduce independently under suitable conditions
Generally much smaller	Generally larger
Lack cytoplasm and cell membrane typical of cells	Have cellular structures including cell membrane and cytoplasm

(d) Three major groups of living organisms — **3 marks**

Any three:

(i) Monera

(ii) Protoctista

(iii) Fungi

(iv) Plantae

(v) Animalia

Total = 10 marks

8.

(a) Characteristics of Monera **(3 Marks):**

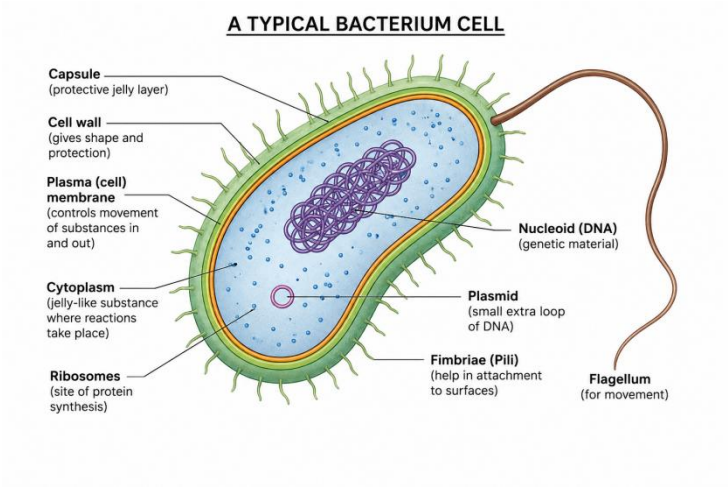
(i) They are prokaryotic (lack a true, membrane-bound nucleus).

(ii) They are single-celled (9unicellular) organisms.

(iii) Their cell walls are composed of peptidoglycan/murein (not cellulose).

(iv) They reproduce mainly by binary fission. *(Any 3 points, 1 mark each)*

(b) Diagram **(6 Marks):**



(c) Roles of Bacteria **(4 Marks):**

(i) *Beneficial:* Used in milk fermentation (yogurt/cheese making); Act as decomposers in ecosystems; Used in sewage treatment; Nitrogen fixation in soil.

(ii) *Harmful:* Cause food spoilage; Cause dangerous diseases in plants and animals.

(d) Disease **(1 Mark):** Cholera, Typhoid, Tuberculosis (TB), or Tetanus.

Total = 10 marks

9. (a) Characteristics of Protoctista **(4.5 Marks):**

(i) They are eukaryotic organisms

(ii) Most are unicellular, though some (like large algae) are multicellular.

(iii) They live mostly in moist or aquatic environments.

(b) Examples **(2 Marks):** Amoeba, Paramecium, Euglena, or Algae/Plasmodium.

(c) Taxonomic Ranks **(3.5 Marks):**

(i) Kingdom

(ii) Phylum (or Division for plants)

(iii) Class

(iv) Order

(v) Family

(vi) Genus

(vii) Species

Total = 10 marks

SECTION C: 15 MARKS

10.

(a) Identification **(1 Mark):** Virus / Viral agent. *(1 Mark)*

(b) Living vs Non-living features **(4 Marks):**

Non-living: It lacks cellular structures (organelles, cytoplasm); It cannot carry out metabolic processes or respiration on its own; It can be crystallised. *Living:* It possesses genetic material (DNA or RNA); It can replicate/reproduce when inside a living host cell.

(c) Binomial Nomenclature Rules **(4 Marks):**

(i) The organism is given two names: the Genus name followed by the Species name.

(ii) The Genus name must always start with a capital letter, while the species name starts with a small letter.

(iii) When printed or typed, the scientific name must be written in *italics*.

(iv) When handwritten, the Genus and Species names must be underlined separately.

(d) Safety Reasons **(6 Marks):**

(i) **Risk of Infection:** Unknown pathogens can easily enter the body through cuts, inhalation, or accidental ingestion, causing severe illness or death.

(ii) **Contamination of the Environment:** Handling specimens without protection can release dangerous microbes into the air or onto laboratory counters, endangering other students.

(iii) **Lack of Immediate Treatment:** Since the micro-organism is unknown, if an accidental infection occurs, there may be no immediate cure, antidote, or vaccine available in the school clinic.

Total = 15 marks